

Array Creation:

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NumPy Analyzer Class
---- Welcome to the NumPy Analyzer! ----
_*_*_*_*_*_*_*_*_*_*_*_*_*_*_*_*_

Choose an option:
1. Create a NumPy Array
2. Perform Mathematical Operations
3. Combine or Split Arrays
4. Search, Sort, or Filter Arrays
5. Compute Aggregates and Statistics
6. Exit
_*_*_*_*_*_*_*_*_*_*_*_*_*_*_*_*_

Enter your choice: 1

Select the type of array to create:
1. 1D Array
2. 2D Array
3. 3D Array
Enter choice: 2
Rows: 3
Cols: 3
Enter 9 elements total:
12 34 56 78 13 35 57 79 24
Array created:
[[12 34 56]
 [78 13 35]
 [57 79 24]]

Choose an operation:
1. Indexing
2. Slicing
3. Back
Enter choice: 1
Enter row index: 2
Enter column index: 1
Element: 79
```

Mathematical Operations:

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Enter your choice: 2
Choose a Mathematical Operation

1. Add
2. Subtract
3. Multiply
4. Divide
5. Dot Product
6. Matrix Multiplication
Enter choice: 5
Enter same-size array elements: 1 2 3 4 5 6 7 8 9
Original Array:
[[12 34 56]
 [78 13 35]
 [57 79 24]]
Second Array:
[[1 2 3]
 [4 5 6]
 [7 8 9]]
Dot Product:
[[540 642 744]
 [375 501 627]
 [541 701 861]]
```

```
Choose a Mathematical Operation

1. Add
2. Subtract
3. Multiply
4. Divide
5. Dot Product
6. Matrix Multiplication

Enter choice: 6
Enter same-size array elements: 1 0 9 7 8 6 5 4 3
Original Array:
[[12 34 56]
 [78 13 35]
 [57 79 24]]

Second Array:
[[1 0 9]
 [7 8 6]
 [5 4 3]]

Matrix Multiplication:
[[ 530  496  480]
 [ 344  244  885]
 [ 730  728 1059]]
```

Combine or Split Arrays:

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Enter your choice: 3
Choose an option:

1. Combine
2. Split
Enter choice: 1
Enter second array elements: 1 2 3 4 5 6 7 8 9
Original Array:
[[12 34 56]
 [78 13 35]
 [57 79 24]]
Second Array:
[[1 2 3]
 [4 5 6]
 [7 8 9]]
After combining:
[[12 34 56]
 [78 13 35]
 [57 79 24]
 [ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]]
=====
```

```
Enter your choice: 3
Choose an option:

1. Combine
2. Split
Enter choice: 2
Enter parts: 3
After splitting:
[array([[12, 34, 56]]), array([[78, 13, 35]]), array([[57, 79, 24]])]
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Search, Sort or Filter Arrays:

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Enter your choice: 4
Choose an option:

1. Search
2. Sort
3. Filter
Enter choice: 1
Enter value: 34
(array([0]), array([1]))
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```
Choose an option:

1. Search
2. Sort
3. Filter
Enter choice: 2
Original Array:
[[12 34 56]
 [78 13 35]
 [57 79 24]]
Array in Ascending order:
[[12 34 56]
 [13 35 78]
 [24 57 79]]
Array in Descending order:
[[24 57 79]
 [13 35 78]
 [12 34 56]]
```

```
Choose an option:

1. Search
2. Sort
3. Filter
Enter choice: 3
Array should be greater than: 50
[56 78 57 79]
=====
```

Compute Aggregates and Statistics:

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Enter your choice: 5
Choose an option:

1. Aggregate
2. Statistics
3. Correlation Coefficient
Enter choice: 1
Sum: 388
Mean: 43.111111111111114
Median: 35.0
Std Dev: 24.15893873885304
Variance: 583.654320987654
```

```
Enter your choice: 5
Choose an option:

1. Aggregate
2. Statistics
3. Correlation Coefficient
Enter choice: 2
Min: 12
Max: 79
Percentile: 50
35.0
```

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Choose an option:
1. Aggregate
2. Statistics
3. Correlation Coefficient
Enter choice: 3
Enter second array: 12 24 46 68 79 35 68 90 11
Correlation:
[[ 1.          -0.6503219  -0.59603956  0.98588696 -0.72057669 -0.69901389]
 [-0.6503219   1.          -0.22235427 -0.5139676  -0.05812177 -0.08865409]
 [-0.59603956 -0.22235427  1.          -0.72205225  0.98624138  0.9908396 ]
 [ 0.98588696 -0.5139676  -0.72205225  1.          -0.82648676 -0.80886662]
 [-0.72057669 -0.05812177  0.98624138 -0.82648676  1.          0.99953136]
 [-0.69901389 -0.08865409  0.9908396  -0.80886662  0.99953136  1.          ]]
```

Exit:

```
Choose an option:
1. Create a NumPy Array
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Enter your choice: 6
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Thank you for using the NumPy Analyzer! Goodbye!
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